Hacettepe University
Computer Engineering Department
BBM233 Logic Design Laboratory
Verilog Final Project
Fall 2025

Deadline: 06/01/2026 at 23:59:59 – NO EXTENSIONS!

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1 World War III: The Dead Hand Protocol

Following decades of geopolitical tension, global powers agreed on one final safeguard against accidental war: an automated escalation control system known as **DEAD-HAND**.

DEAD-HAND is not a weapon. It is a *decision machine*.

Its sole purpose is to monitor global military activity, evaluate threat levels, and coordinate defensive and retaliatory readiness. Human commanders remain in the loop — but only up to a point.

Once DEAD-HAND enters its final escalation sequence, **World War III becomes inevitable**.

Your mission: Model DEAD-HAND's control logic in Verilog, simulate its behavior, and determine whether escalation can be stopped **before the Point-of-No-Return**.

2 System Specifications

2.1 I/O Ports (Do NOT change)

You MUST use the provided starter code. Do NOT change I/O names, order, or bit widths.

2.2 Inputs

Signal	Width	Description
clk	1	1 Hz clock (1 tick per second)
reset	1	Synchronous reset (active-high)
threat_level	2	00=Peace, 01=Suspicious, 10=Hostile, 11=Nuclear
diplomatic_override	1	Emergency override attempt
comms_lost	1	Communication blackout
system_fault	1	Internal system failure

3 Main FSM States

State	Code	Meaning
PEACE	000	Normal monitoring
ALERT	001	Suspicious activity
MOBILIZATION	010	Forces preparing
ENGAGEMENT	011	Active planning (Sub-FSM active)
GLOBAL_WAR	101	Full-scale war (final)
DEADLOCK	110	Point-of-No-Return (final)

4 Engagement Sub-FSM States

State	Code	Meaning
ARM	00	Forces armed
TRACK	01	Target tracking
AUTHORIZE	10	Launch authorization
ABORT	11	De-escalation

5 Behavior Rules

5.1 Global Immediate Rules

R1: reset=1 → PEACE, Sub-FSM=ABORT, timers cleared

R2: system_fault=1 in non-final state → GLOBAL_WAR

R3: DEADLOCK and GLOBAL_WAR are irreversible

5.2 ENGAGEMENT Exit Policy

- AUTHORIZE for 3 seconds → DEADLOCK
- ABORT for 5 seconds → MOBILIZATION

Further de-escalation depends on threat levels after MOBILIZATION.

5.3 Late Override Rule

override_ignored is set if diplomatic override is asserted:

- in DEADLOCK or GLOBAL_WAR, or
- after 2 seconds in AUTHORIZE

Once set, it remains high until reset.

6 Threat Level and Communication-Based Transitions Explained

This section provides a detailed explanation of how `threat_level` and `comms_lost` signals influence state transitions in the Main FSM. These explanations are intended to clarify the temporal and priority-based behavior required by the specification.

6.1 Threat Level Evaluation Model

The `threat_level` input represents the globally observed military posture. It is evaluated **once per second** on each rising edge of the clock.

> Threat-based transitions require the specified condition to hold **continuously for the given number of seconds**. If the condition is violated at any second, the associated timer or counter must be reset.

- Transitions such as `PEACE` → `ALERT` and `ALERT` → `MOBILIZATION` are **time-qualified**.
- The system must count consecutive seconds during which the required threat condition holds.
- A single second of violation immediately resets the counter.

6.2 Threat Level Transition Summary

Current State	Condition	Resulting Transition
PEACE	`threat_level` $\geq$ `01 for 5s`	Transition to ALERT
ALERT	`threat_level` $\geq$ `10 for 10s`	Transition to MOBILIZATION
ALERT	`threat_level = 00 for 4s`	Return to PEACE
MOBILIZATION	`threat_level = 11 or comms_lost = 1`	Transition to ENGAGEMENT
MOBILIZATION	`threat_level` $\leq$ `01 for 4s`	De-escalation to ALERT

6.3 Role of `comms_lost`

The `comms_lost` signal represents a complete communication blackout between the DEAD-HAND system and human command authority.

> Loss of communication is treated as an extreme risk factor and may trigger immediate escalation independent of the threat level.

- In `MOBILIZATION`, if `comms_lost=1`, the system transitions immediately to `ENGAGEMENT`, regardless of `threat_level`.
- Once in `ENGAGEMENT`, `comms_lost` has no further effect on Main FSM transitions; control is delegated to the Sub-FSM.

6.4 Priority and Conflict Resolution

If multiple transition conditions are true simultaneously, the following priority order must be respected:

P1: `reset`

P2: `system_fault`

P3: Final states (`DEADLOCK`, `GLOBAL_WAR`)

P4: Immediate escalation conditions

P5: Time-qualified threat-based transitions

This priority ordering ensures deterministic behavior and prevents ambiguous state transitions.

Students are strongly encouraged to implement explicit counters or timers for each time-qualified transition to ensure correctness under automatic grading.

6.5 Sub-FSM Transition Rules (Active During ENGAGEMENT)

Sub-FSM is active only while Main FSM is in `ENGAGEMENT`. When Main FSM is not in `ENGAGEMENT`, Sub-FSM must be held/reset to `ABORT` and its timer cleared.

S1: Entry to ENGAGEMENT: Sub-FSM starts at `ARM`, sub-timer resets to 0.

S2: ARM → TRACK: After 4 seconds in `ARM`, if no diplomatic override occurs.

S3: TRACK → AUTHORIZE: After 6 seconds in `TRACK`, if no diplomatic override occurs.

S4: Any Sub-State → ABORT: If diplomatic_override=1 while Main FSM is not in `DEADLOCK` and not in `GLOBAL_WAR`.

S5: AUTHORIZE hold: If Sub-FSM stays in `AUTHORIZE` for 3 seconds, Main FSM transitions to `DEADLOCK`.

S6: ABORT hold: If Sub-FSM stays in `ABORT` for 5 seconds, Main FSM transitions back to `MOBILIZATION`.

7 Example Waveforms

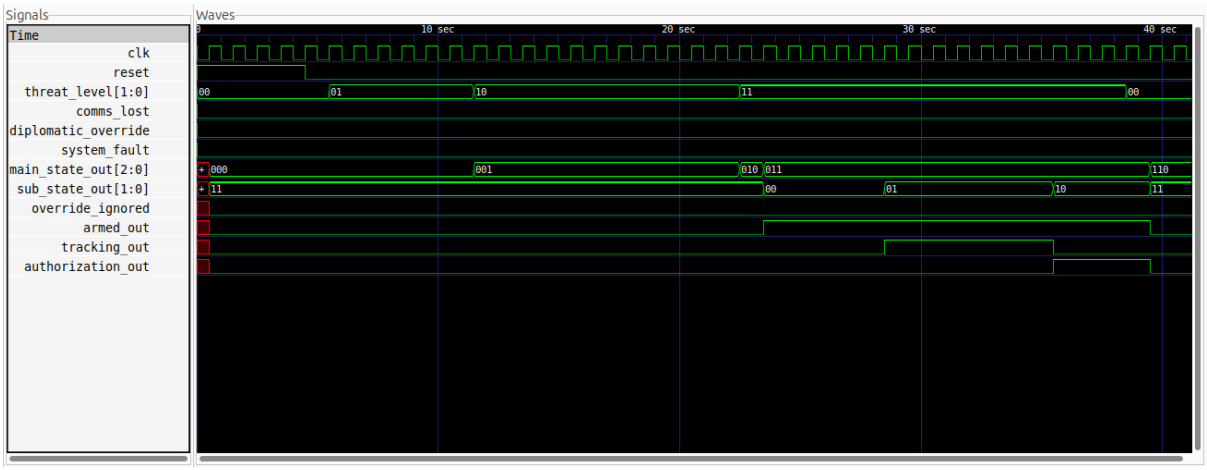

Figure 1: Scenario 1: Normal escalation to DEADLOCK

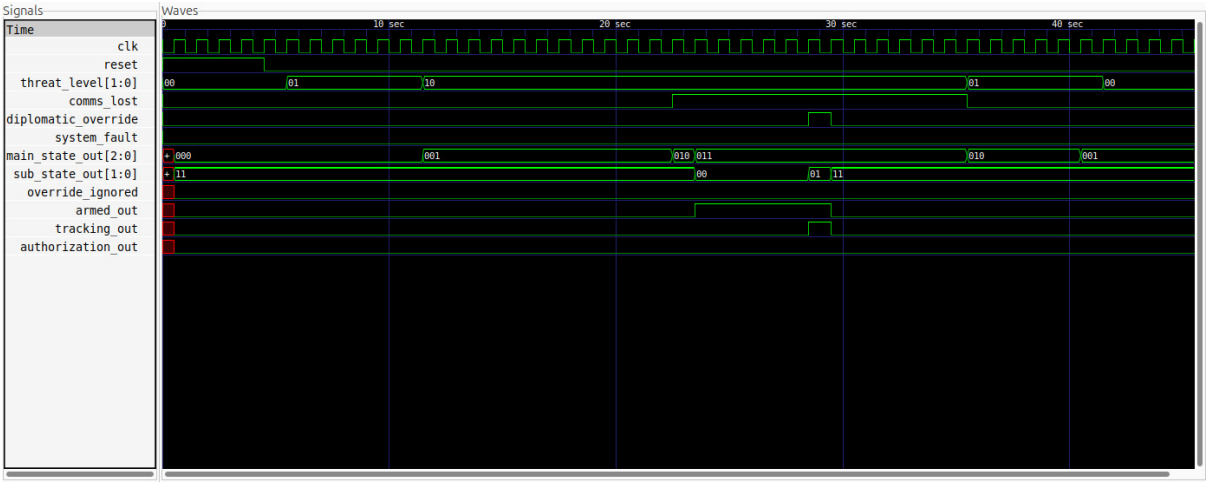

Figure 2: Scenario 2: Successful ABORT and de-escalation

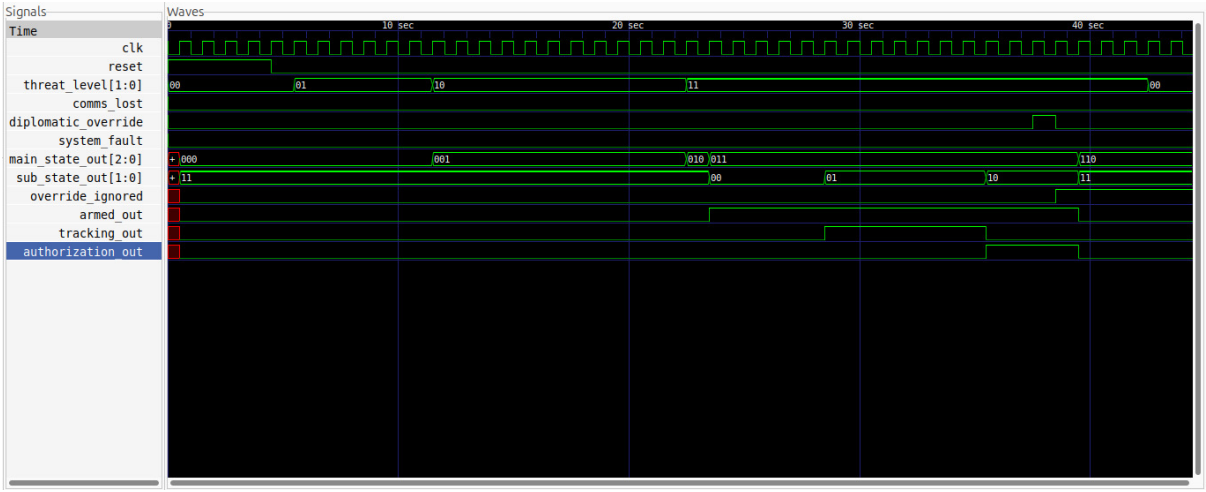

Figure 3: Scenario 3: Late override ignored

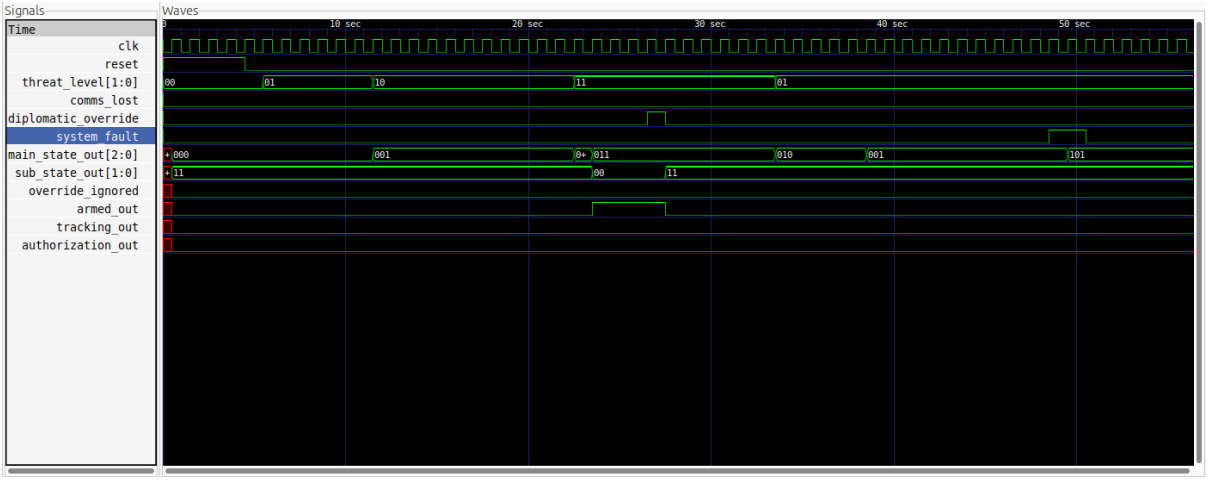

Figure 4: Scenario 4: Abort, de-escalation, then system fault

8 Starter Code

Students are provided with a starter Verilog file containing the correct module interface and state encodings.

Modifying the module interface or state encodings is strictly forbidden.

9 Submission Rules

- Students must submit their solutions as a **ZIP archive**.
- The ZIP file must be named exactly:

```
b<studentID>.zip
```

- The archive must contain **only one file**:
  - `dead_hand.v`

The automatic grading system relies on a strict file structure. Any deviation will cause the submission to be rejected automatically.

10 Academic Integrity and Plagiarism

This project must be completed **individually**.

- Sharing code, receiving code, or using another student's solution is strictly prohibited.
- Using publicly available solutions or generative AI tools to directly produce Verilog code for this project is forbidden.
- Students are responsible for understanding and being able to explain every line of their submitted code.

All submissions will be checked using automated similarity detection tools. Cases of suspected plagiarism will be reported in accordance with Hacettepe University academic integrity regulations.

Academic misconduct may result in a failing grade for the assignment and further disciplinary action.

11 Important Notes

- **Synchronous Reset:** Initial X values before the first rising clock edge are expected and not considered an error.
- System operates strictly in seconds.
- Automatic grading is waveform-based and exact.

Good luck. Try not to start World War III.